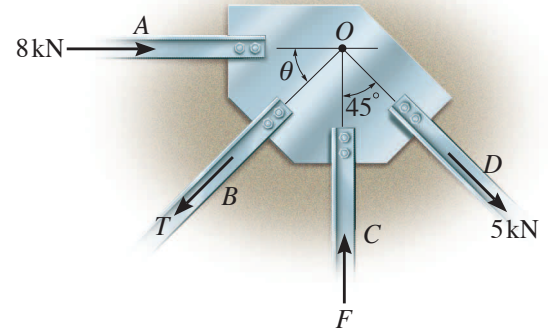


The problems in this supplement are from Chapter 3 of “Engineering Mechanics: Statics” 13th Ed. by R.C. Hibbeler 2013.

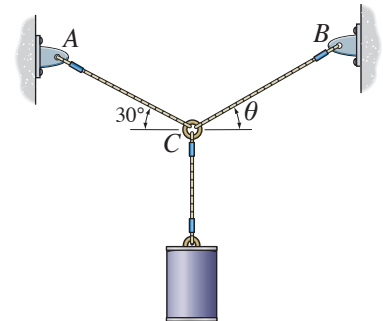
S-6.

The gusset plate is subjected to the forces of four members. Determine the force in member B and its proper orientation θ for equilibrium. The forces are concurrent at point O . Take $F = 12 \text{ kN}$.



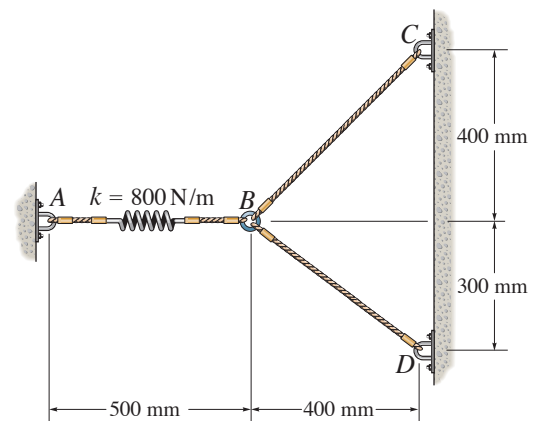
S-10

Determine the tension developed in wires CA and CB required for equilibrium of the 10-kg cylinder. Take $\theta = 40^\circ$.



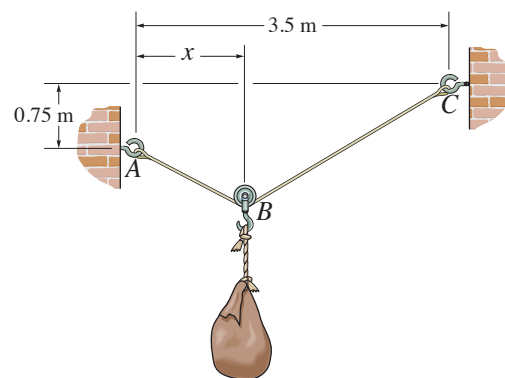
S-15

The spring has a stiffness of $k = 800 \text{ N/m}$ and an unstretched length of 200 mm . Determine the force in cables BC and BD when the spring is held in the position shown.



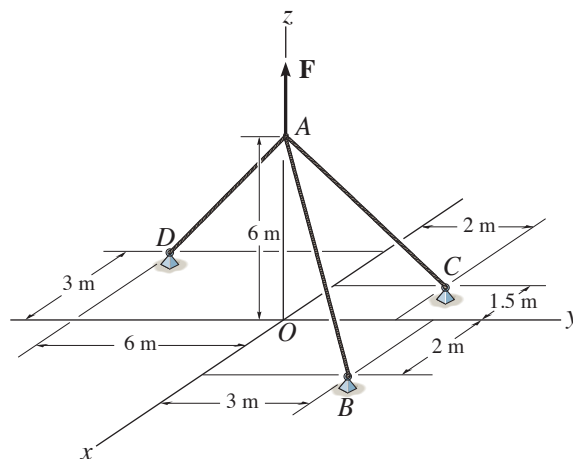
S-35

Cable ABC has a length of 5 m. Determine the position x and the tension developed in ABC required for equilibrium of the 100-kg sack. Neglect the size of the pulley at B .



S-44

If cable AB is subjected to a tension of 700 N, determine the tension in cables AC and AD and the magnitude of the vertical force $\mathbf{F}$.



S-54

Determine the tension developed in cables AB and AC and the force developed along strut AD for equilibrium of the 400-lb crate.

